

- 4 (a) Electric field strength at a point is defined as the electric force per unit positive charge that acts on a stationary charge placed at that point.

$$E = F/q$$

N C⁻¹ or V m⁻¹ (question asked for SI units. NOT base SI units)

- (b) (i) (vertically) downwards
(or top to bottom, or down the page)

(ii) Force on sphere $F = kx = 0.24 \times 18 \times 10^{-3}$
 $= 4.32 \times 10^{-3}$

(iii) $E = F/q = 4.32 \times 10^{-3} / 4.1 \times 10^{-8}$
 $E = V/d = (4.32 \times 10^{-3} / 4.1 \times 10^{-8}) = 5.0 \times 10^3 / d$
 $d = 4.76 \times 10^{-2}$

- (c) Electric field becomes zero (or ceases to exist)
Flow of charge (or electrons) from one plate to the other (or plates discharge)
Potential difference across plates becomes zero (or no pd across plates, or plates at same potential)